

Security incident report

Section 1: Identify the network protocol involved in the incident

- **TCP (Transmission Control Protocol)** – Used to establish connections between the client (your machine) and the web servers using the **three-way handshake (SYN, SYN-ACK, ACK)**.
- **HTTP (Hypertext Transfer Protocol)** – Used after the connection is established to send the **GET / request** to retrieve web content over port 80.
- **DNS (Domain Name System)** – Used to translate the domain name **greatrecipesforme.com** into an IP address (**192.0.2.17**).

Section 2: Document the incident

- The system first sends HTTP requests to **yummyrecipesforme.com**, completing normal TCP handshakes and GET requests over port 80.
- Later, a DNS query is made for **greatrecipesforme.com**, which resolves successfully to **192.0.2.17**.
- Immediately after, the machine initiates multiple TCP connections (SYN packets) to the new server on port 80.
- The logs show repeated connection attempts followed by large amounts of traffic on port 80 (“a lot of traffic”), suggesting abnormal or excessive requests.
- This pattern may indicate a **high-volume automated attack**, such as a **brute force attempt or HTTP-based flooding behavior**, where repeated requests overwhelm the server.

The incident occurred within network traffic logs showing communication between a local machine (“your.machine”) and multiple external web servers, including **yummyrecipesforme.com** and **greatrecipesforme.com**. The

activity was captured over standard web traffic on **port 80 (HTTP)** and includes both DNS and TCP connection data.

The sequence of events began at approximately **14:18:36**, when the local machine initiated a TCP connection to **yummyrecipesforme.com** using a SYN packet. The server responded with a SYN-ACK, and the connection was successfully established through the final ACK step of the TCP three-way handshake. Immediately after the connection was established, the client sent an HTTP **GET /** request to retrieve the homepage. The server acknowledged the request, and additional traffic followed on port 80, indicating continued data exchange.

At **14:20:32**, the local machine sent a DNS query to **dns.google** requesting the IP address for **greatrecipesforme.com**. The DNS server responded with an A record resolving the domain to **192.0.2.17**. This resolution allowed the system to establish a connection with the second domain.

At **14:25:29**, multiple TCP connection attempts were initiated from the local machine to **greatrecipesforme.com** over port 80. The logs show repeated SYN, SYN-ACK, and ACK sequences, indicating multiple successful connection establishments. Each connection was followed by an HTTP GET request for the root resource. After these requests, the log indicates a significant increase in traffic volume on port 80, described as “a lot of traffic,” suggesting sustained or repeated automated requests.

The traffic pattern observed includes repeated connection establishment and repeated HTTP requests within a short time window. Evidence for this conclusion is based on packet capture logs showing:

- TCP SYN/SYN-ACK/ACK handshake sequences
- DNS resolution responses from **dns.google**
- HTTP GET requests to both domains
- Increased frequency of connections and sustained traffic on port 80

No direct user interaction or manual confirmation events are recorded in the logs provided. The activity appears to be automatically generated based on the consistent timing and repetition of connection attempts.

Section 3: Recommend one remediation for brute force attacks

One effective remediation is to implement **rate limiting**.

- Rate limiting restricts how many requests a single IP address can make

in a given time period.

- This helps prevent attackers from repeatedly trying many login attempts or sending excessive requests.
- It reduces server overload and blocks automated brute force tools from successfully overwhelming the system.

A recommended security measure is to **limit the number of login attempts (account lockout policy)**.

This control is effective because it prevents repeated automated attempts to guess passwords. After a defined number of failed login attempts, the system temporarily locks the account or delays additional attempts. This significantly reduces the speed and success rate of brute force attacks by stopping attackers from continuously trying different password combinations. It also provides time for security systems to detect and respond to suspicious login behavior.